

THERMOPHYSICAL PROPERTIES OF BINARY FLUORINATED FLUIDS: NUMERICAL STUDY OF FC-72/FC-770 AND FC-72/HFE-649 MIXTURES

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ABSTRACT

The development of compact heat exchangers requires working fluids with tunable thermophysical properties. Fluorinated engineered fluids such as FC-72, FC-770, and HFE-649 are chemically stable, dielectric, and widely used in electronics cooling. However, the fixed properties of pure fluids often limit performance, motivating the use of binary mixtures as adjustable working media. This study examines two systems, FC-72/FC-770 and FC-72/HFE-649, over the full composition range (mass fractions 100/0, 75/25, 50/50, 25/75, and 0/100). Density, kinematic viscosity, thermal conductivity, specific heat capacity, and atmospheric-pressure boiling temperature were measured experimentally, analyzed as functions of mixture composition, and used as CFD input data. For FC-72/FC-770 mixtures, density increased steadily from 1680 kg/m³ (pure FC-72) to 1793 kg/m³ (pure FC-770), while kinematic viscosity also increased, indicating higher flow resistance in FC-770-rich blends. Thermal conductivity varied nonlinearly, with a minimum at intermediate compositions (~0.049–0.052 W/m·K) and a maximum of 0.063 W/m·K for pure FC-770. Specific heat decreased from 1100 J/kg·K (FC-72) to ~0.93 J/kg·K at 50/50, then increased toward 1.04 J/kg·K (FC-770). The boiling temperature shifted from about 329 K (FC-72) to 368 K (FC-770). For FC-72/HFE-649 mixtures, the density trend reversed, decreasing from 1680 kg/m³ (FC-72) to 1600 kg/m³ (HFE-649). Thermal conductivity showed limited variation (0.048–0.059 W/m·K), while specific heat increased with HFE-649 content, reaching 1.10 J/kg·K for the pure fluid. The boiling temperature decreased from 329 K (FC-72) to 322 K (HFE-649). Numerical simulations of flow and heat transfer in a vertical multichannel module were performed in STAR-CCM+. The model considered laminar upward flow in rectangular channels heated by a Haynes 230 foil. Although the detailed CFD results will be reported separately, the preliminary analysis indicates that mixture composition provides a practical lever for tuning saturation margin and hydraulic response; however, intermediate blends may not deliver a systematic enhancement of single-phase convective heat transfer relative to the pure fluids under the conditions studied. Future work will extend the analysis to performance metrics that couple heat transfer with pressure losses and will further refine the operating window for compact heat-exchanger applications.

SELECTED BINARY MIXTURES AND PROPERTY ANALYSIS

Two fully miscible binary systems were selected for the present study: FC-72/FC-770 and FC-72/HFE-649. All three fluids are dielectric, chemically stable fluorinated engineered liquids used in thermal management, and blending them with FC-72 enables composition-based tuning of key properties relevant to compact heat exchangers (density, kinematic viscosity, thermal conductivity, and specific heat) while maintaining practical compatibility with typical pumping and construction requirements. Mixtures were prepared at mass fractions of 100/0, 75/25, 50/50, 25/75, and 0/100 (FC-72/second component), and the resulting properties were evaluated and used as input data for the numerical analysis.

Table 2 Binary mixtures FC-72/FC-770 and FC-72/HFE-649: components, mass fractions, and selected physical properties

Mixture	FC-72/ FC-770	FC-72/ FC-770	FC-72/ FC-770	FC-72/ HFE-649	FC-72/ HFE-649	FC-72/ HFE-649
Mass Fractions	25/75	50/50	75/50	25/75	50/50	75/50
Density, kg/m ³	1778	1753	1727	1638	1657	1684
Kinematic viscosity, mm ² /s	1.094	0.607	0.536	0.457	0.461	0.522
Thermal conductivity, W/(m·K)	0.052	0.050	0.049	0.049	0.048	0.049
Specific heat, J/(kg·K)	0.933	0.937	0.941	0.957	0.940	0.923
Boiling temperature $T_{sat,theor}/T_{sat,exp}$, K	329 / 341	329 / 339	329 / 337	322 / 327	322 / 327	322 / 327

Table 1 Selected physical properties of FC-72, FC-770 and HFE-649

Fluid	FC-72	FC-770	HFE-649
Density, kg/m ³	1680	1793	1600
Kinematic viscosity, mm ² /s	0.38	0.79	0.40
Thermal conductivity, W/(m·K)	0.057	0.063	0.059
Specific heat, J/(kg·K)	1.1	1.038	1.103
Boiling temperature, K	329	368	322

Three binary mixtures were selected for each system, containing 25%, 50% and 75% mass fractions of the second component, with the remainder being FC-72 (i.e., FC-72/FC-770 and FC-72/HFE-649 at 75/25, 50/50 and 25/75). For each mixture, density, kinematic viscosity, dynamic viscosity, thermal conductivity and specific heat capacity were measured experimentally, and the resulting values are reported for the chosen reference conditions of the property calculations. In addition, normal boiling temperatures of the pure fluids were taken as atmospheric-pressure reference points. Table 2 summarizes the determined properties of the investigated binary mixtures.

EXPERIMENTAL SETUP

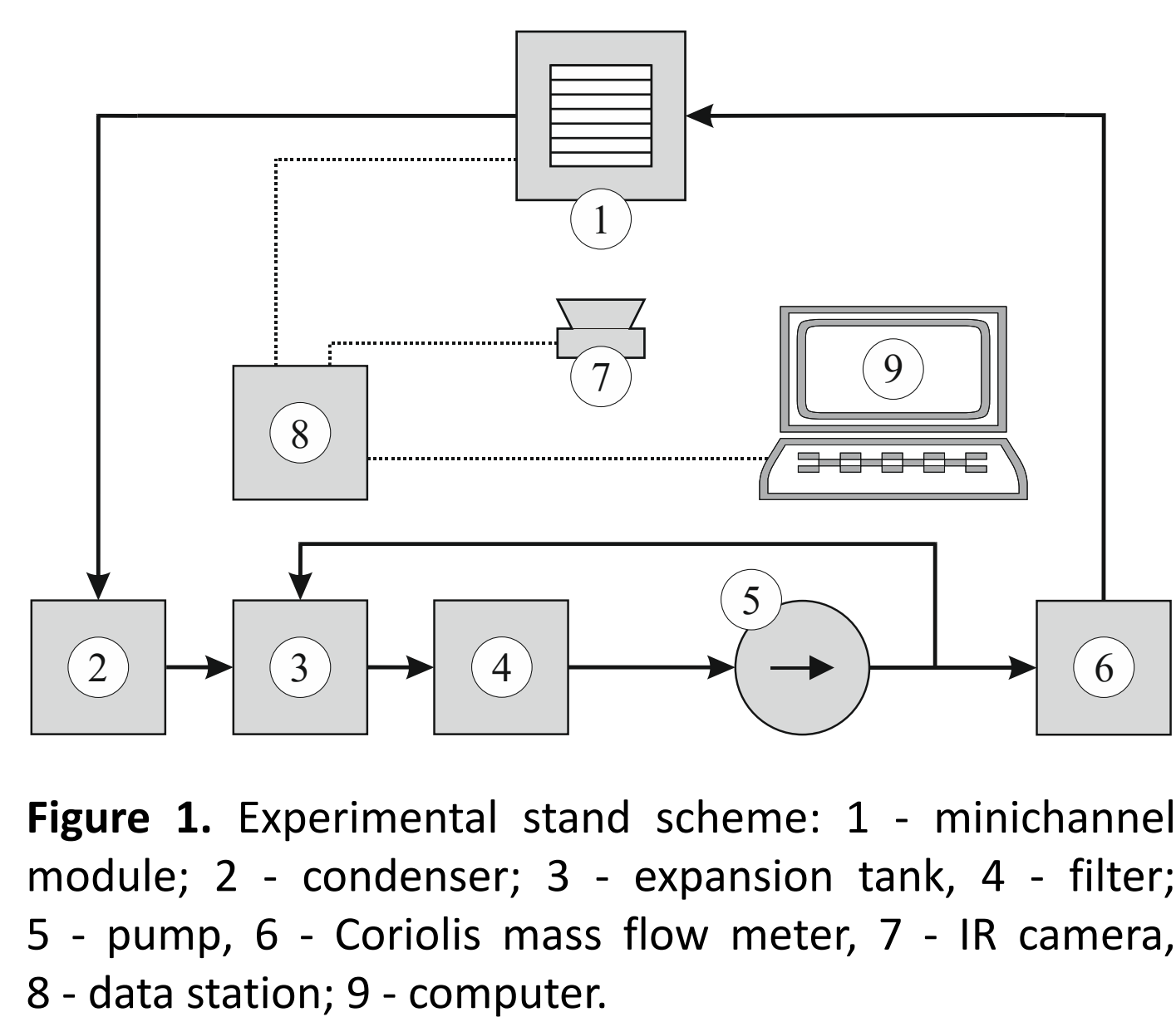


Figure 1. Experimental stand scheme: 1 - minichannel module; 2 - condenser; 3 - expansion tank, 4 - filter; 5 - pump, 6 - Coriolis mass flow meter, 7 - IR camera, 8 - data station; 9 - computer.

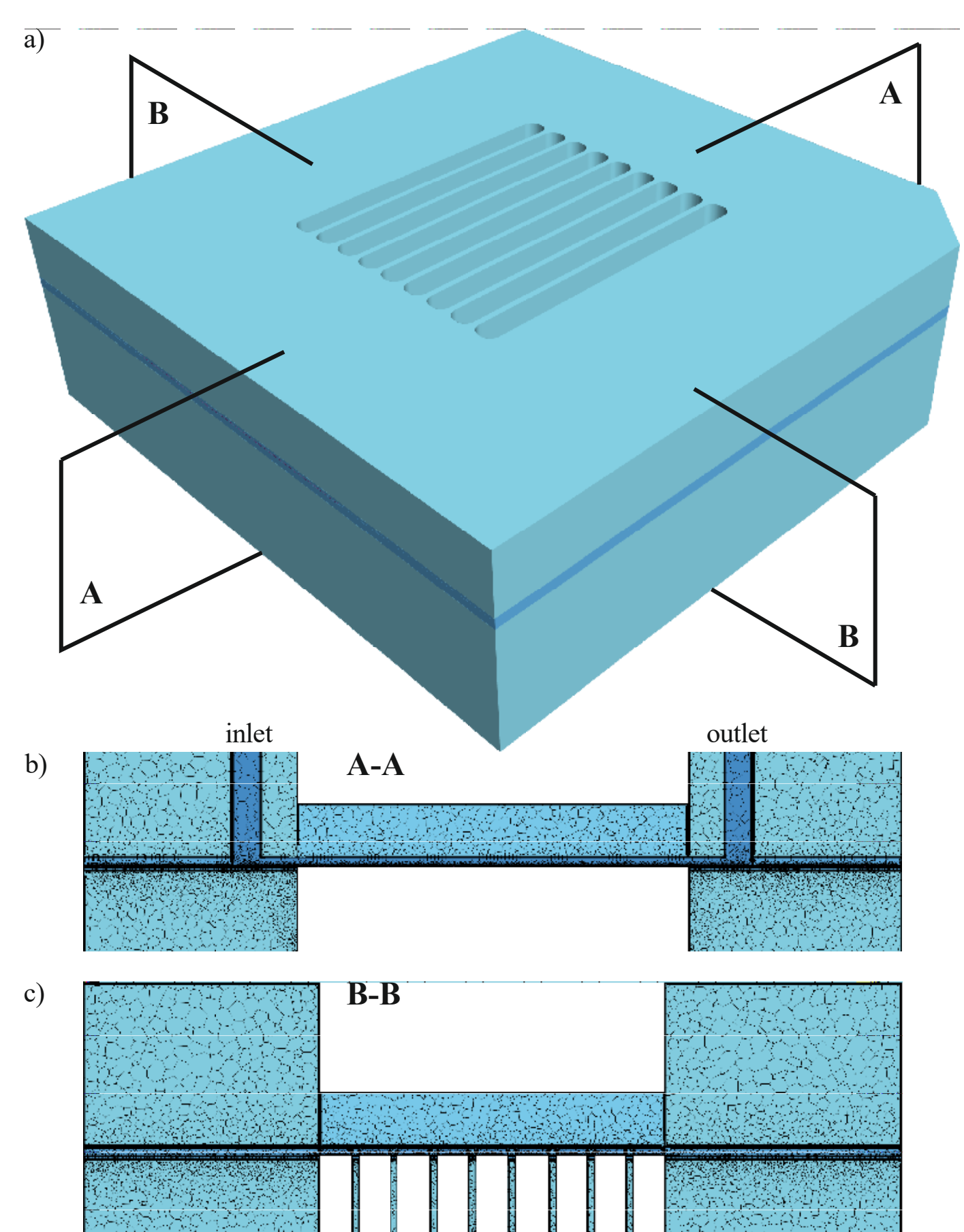


Figure 2. Schematic of the experimental module: (a) view, (b) longitudinal cross-section, (c) transverse cross-section.

RESULTS AND ANALYSIS

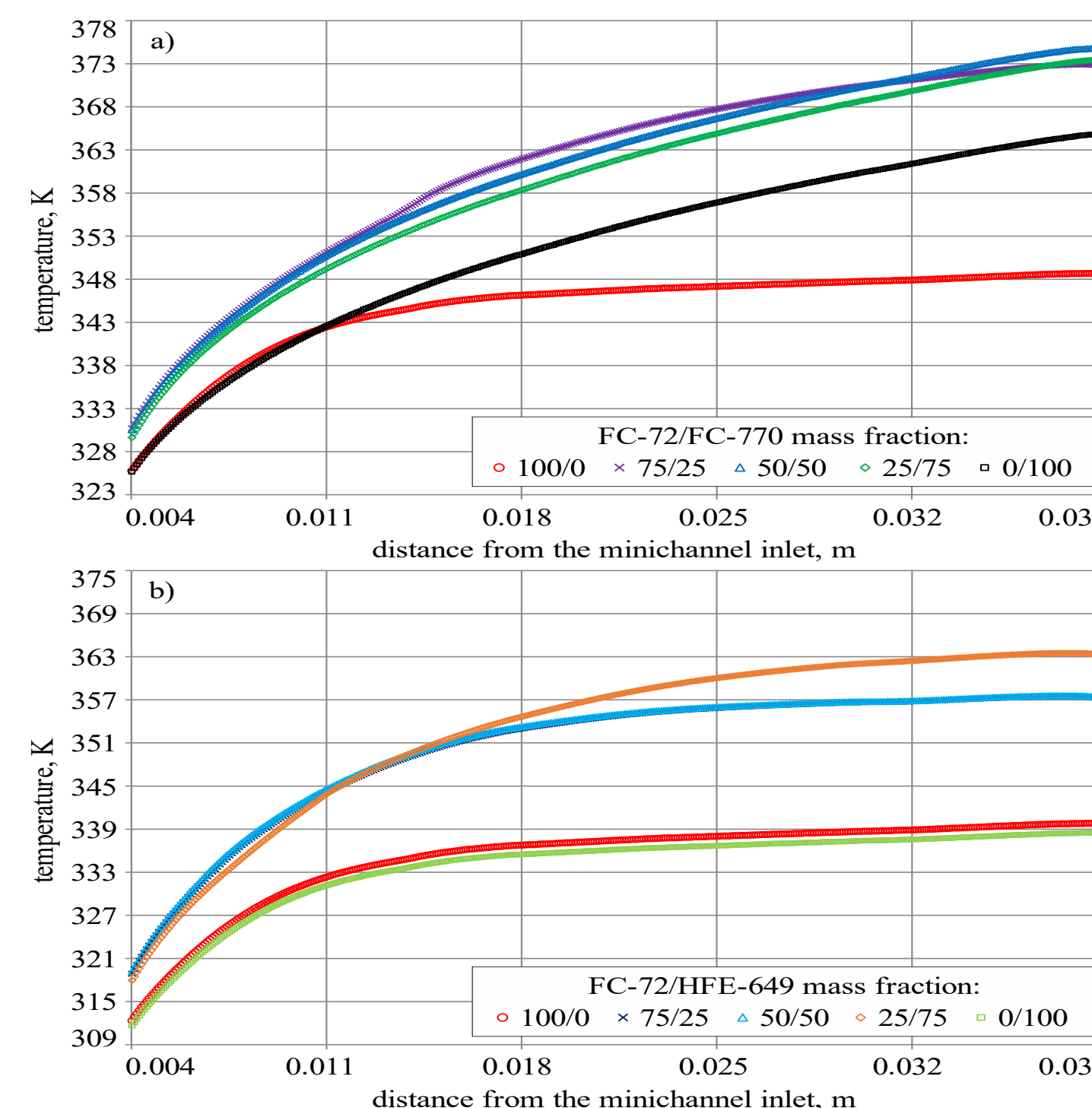


Figure 3. Axial temperature distributions along the minichannel for: a) FC-72/FC-770, b) FC-72/HFE-649 mixtures

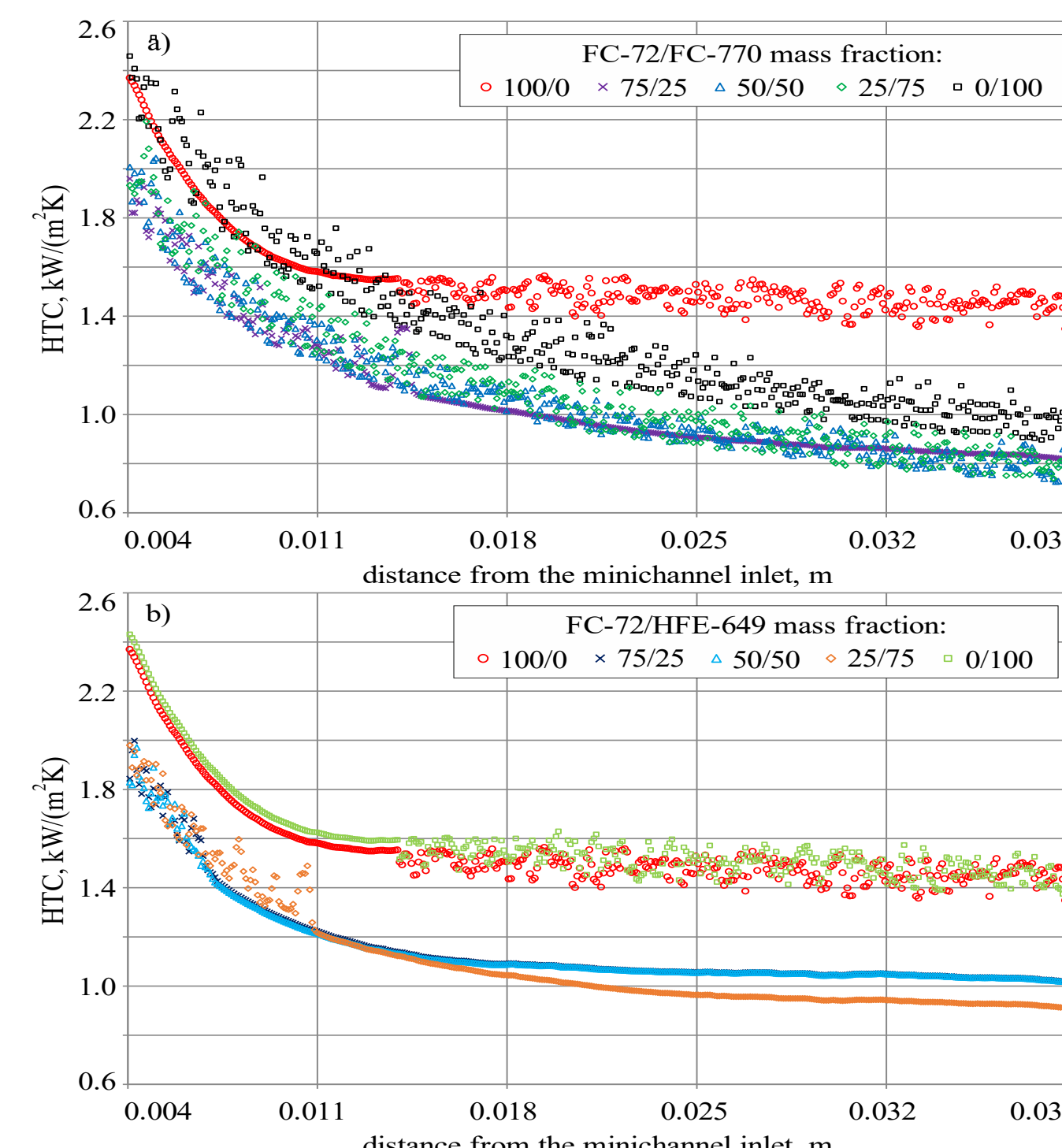


Figure 4. Axial HTC distributions along the minichannel for: a) FC-72/FC-770, b) FC-72/HFE-649 mixtures

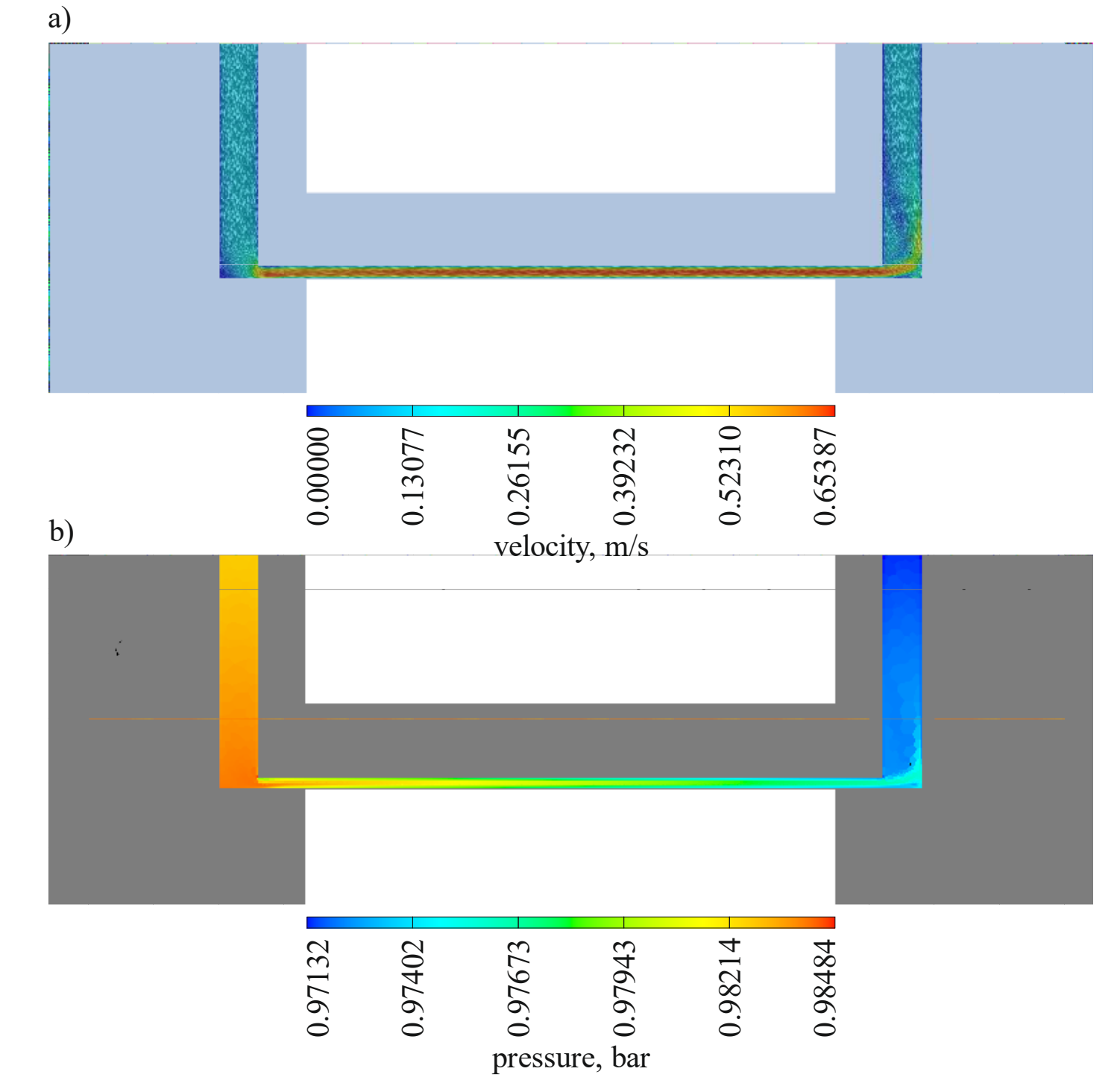


Figure 5. Representative CFD results in the minichannel module (longitudinal section) for a 50/50 FC-72/FC-770 mixture: (a) velocity magnitude field, (b) static pressure field.

CONCLUSIONS

- ✓ Two binary dielectric fluorinated systems, FC-72/FC-770 and FC-72/HFE-649, were evaluated over composition ranges relevant to compact minichannel heat exchangers, using experimentally established thermophysical-property inputs.
- ✓ The mixture-property trends are clearly non-ideal. In both systems, thermal conductivity exhibits a minimum at intermediate mass fractions, and kinematic viscosity shows excess behavior, with several blends exceeding the viscosities of the pure components (most pronounced in FC-72/FC-770).
- ✓ Three-dimensional conjugate heat-transfer simulations in the vertical multichannel module reproduce the expected signature of thermally developing laminar single-phase flow: high HTC values in the inlet region followed by a monotonic downstream decay toward quasi-asymptotic levels, with distinct inlet and outlet end effects.
- ✓ Under the investigated operating conditions, intermediate blends do not provide a systematic single-phase heat-transfer advantage over the pure fluids; in multiple cases, they produce lower HTC levels along most of the channel length, which is also reflected by higher axial temperature levels for the same heating input.
- ✓ The HTC penalties are consistent with the coupled property - flow response: reduced thermal conductivity lowers diffusive heat transport, while elevated viscosity increases hydraulic resistance and modifies the velocity/Reynolds-number field, promoting earlier thermal development and reducing local HTC.
- ✓ The primary value of these results is therefore not the demonstration of universal heat-transfer enhancement, but a quantified property - performance map that identifies composition ranges that are unfavorable for single-phase operation in the studied geometry and boundary-condition set.
- ✓ From an application standpoint, binary mixtures remain useful as tuning media because composition control can shift the effective saturation margin (upward with FC-770 and downward with HFE-649) and adjust density/viscosity for constraints such as allowable pressure drop, flow distribution across parallel channels, and temperature/pressure operating windows in dielectric cooling loops; subsequent analysis should therefore consider combined metrics that couple heat transfer with pumping power rather than HTC alone.

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